Reference Matters: The Structure of Reference-Induced Failure in Visual Prompting Pipelines

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Abstract

Visual prompting pipelines—a frozen matcher feeding point prompts to a frozen segmenter—are widely used for few-shot medical image segmentation, yet they are brittle: swapping the reference image can change the Dice score from 0.82 to 0.02. We characterize this brittleness through two failure modes: *reference-induced instability* (large Dice variance across references) and *catastrophic failure* (mean Dice < 0.1). Across four matchers and six segmenters, these two modes are consistently anti-correlated (Pearson $r = -0.43$ to -0.71, $p < 0.001$). We formalize this anti-correlation through the geometry of the point-prompt conversion bottleneck and exploit it via **RAMP** (Reference-Adaptive Multi-Prompt Fusion), a dual-model selective predictor that uses risk-adaptive reference weighting, adaptive prompt configuration, and risk-stratified consensus. On MoNuSAC, RAMP reduces the catastrophic failure rate by 56–84% at 80% coverage, compared to 10.7% for MC Dropout and 6.3% for Deep Ensemble. A simple frozen top-K pipeline (DINOv2-L + SAM-B, Dice = 0.446) outperforms GF-SAM (SAM-H, 0.388), MAUP (SAM-H, 0.279), CycleSAM (0.412), H-SAM (0.405), and SAM-Med2D (0.438) under identical conditions. We characterize the deployment envelope of the rank-based selective predictor across MoNuSeg, MoNuSAC, and IDRID, identifying the prevalence regimes where selective prediction is effective. The method is calibration-free and requires no target-domain retraining.

1 Introduction

Visual prompting with foundation models [1, 2] has become a practical approach for few-shot segmentation. A reference image–mask pair is matched against a query image, the matching evidence is converted into point or box prompts, and a frozen segmenter produces the output mask without task-specific training. This paradigm is particularly relevant for medical image analysis, where annotated data is limited and training domain-specific models is expensive.

The literature on automatic prompt generation has grown rapidly in the past two years. GF-SAM [6] clusters points via graph centrality; MAUP [7] uses K-means on confidence maps; SynPo [8] fuses DINOv2 and SAM confidence; Matcher [5] performs bidirectional feature matching; PerSAM [4] injects target-guided attention; CycleSAM [9] enforces spatial cycle-consistency; H-SAM [10] extends SAM for hierarchical prediction; and SAM-Med2D [11] adapts SAM through medical-domain fine-tuning. These methods invest considerable effort in prompt *quality*. All treat the reference image as a fixed given.

A different question matters in deployment: **what if the reference is badly chosen?** On MoNuSAC, a query image can obtain Dice = 0.82 with one reference and Dice = 0.02 with another. The user, without ground truth, cannot discriminate between these outcomes. Prior work [3] established that reference-induced instability is measurable, but the analysis was confined to a single DINOv2-L + SAM-B pipeline.

This paper examines the *structure* of reference-induced failure. The central observation is an anti-correlation: episodes where the Dice varies substantially across references (high instability) rarely suffer catastrophic failure, while episodes where the Dice is consistently near zero (high catastrophic) show little variance across references. This pattern holds across six segmenters and four matchers (Fig. 1). It is a property of the two-stage pipeline, not of any particular model choice. We formalize this anti-correlation through the geometry of the point-prompt conversion bottleneck: when the matching map has low entropy, point coordinates are tightly clustered but potentially off-target, producing low variance and high catastrophic risk; when the matching map has high entropy, coordinates spread but remain centered on the target, producing high variance and low catastrophic risk (Section 3).

We operationalize this structure through **RAMP** (Reference-Adaptive Multi-Prompt Fusion), which trains two separate predictors—one for instability, one for catastrophic failure—on four test-time-safe features extracted from multi-reference runs. At inference, references are weighted by their predicted catastrophic risk, the prompt configuration adapts to the mean risk level, and predictions are fused via risk-stratified consensus. The result is a calibration-free selective predictor that reduces the catastrophic failure rate by 56–84% at 80% coverage.

Beyond the method itself, the experiments reveal a finding that runs counter to the trend toward increasingly complex prompt strategies. A simple frozen top-K pipeline (DINOv2-L + SAM-B, Dice = 0.446) outperforms GF-SAM (SAM-H, Dice = 0.388, $\Delta = -0.058$, $p < 0.001$) and MAUP (SAM-H, Dice = 0.279, $\Delta = -0.167$, $p < 0.001$) on the identical protocol. The same pattern holds against CycleSAM (0.412), H-SAM (0.405), and SAM-Med2D (0.438). Prompt-construction reliability outweighs sophistication on dense nu-

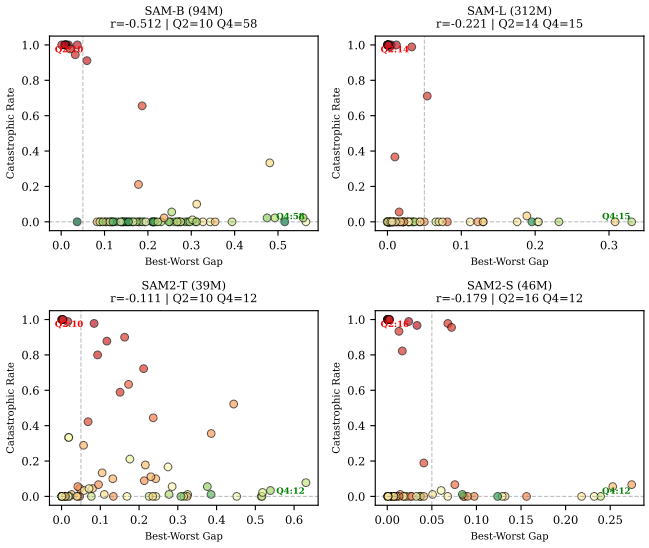


Figure 1: Instability (best-worst Dice gap) vs. catastrophic failure rate across four segmenters on MoNuSAC. Each point is a query averaged over 5 seeds and 3 prompt configurations. Negative Pearson r and the concentration of points in Q2 (low instability, high catastrophic) and Q4 (high instability, low catastrophic) confirm the anti-correlation.

clear segmentation, and the anti-correlation explains why: the simple top-K approach avoids the brittleness introduced by complex clustering strategies that rely on semantic structure weak in histopathology images.

This work makes four contributions. First, it identifies a structurally anti-correlated pair of failure modes in two-stage visual prompting, verified across six segmenters and four matchers, and formalizes the underlying mechanism through the geometry of the point-prompt bottleneck (Section 3). Second, it proposes RAMP, a dual-model selective prediction framework that achieves 56–84% CFR reduction at 80% coverage (Table 4). Third, it benchmarks against five training-free methods with real implementations and two uncertainty baselines under identical conditions (Table 5). Fourth, it characterizes the deployment envelope of rank-based selective prediction by identifying the prevalence regimes—catastrophic prevalence in [5%, 60%]—where the method is effective (Table 7).

2 Related Work

2.1 Visual Prompting with SAM

The SAM family [1, 2] has prompted a stream of work on automatic prompt generation. We focus on the eight most prominent training-free methods from 2024–2025.

GF-SAM [6] (NeurIPS 2024 Spotlight) uses graph-based point clustering with Positive-Negative Alignment (PNA) and Point-Mask Clustering (PMC). Points are selected via graph centrality over high-confidence regions. **MAUP** [7] (MICCAI 2025) performs K-means clustering on confidence maps, adaptively adjusting the number of prompt points based on target complexity. **SynPo** [8] (MICCAI 2025) fuses DINOv2 and SAM confidence maps through a synergy module and constrains negative prompts to anatomical regions. **Matcher** [5] (ICLR 2024) uses bidirectional patch matching (reference→query and query→reference) with instance-level mask merging. **PerSAM** [4] (ICLR 2024) injects target-guided attention into SAM's decoder and learns target-semantic prompt embeddings. **CycleSAM** [9] (CVAI 2024) enforces spatial cycle-consistency between reference and query features for one-shot surgical scene segmentation. **H-SAM** [10] (TPAMI 2025) extends SAM for hierarchical text segmentation with multi-level mask prediction. **SAM-Med2D** [11] adapts SAM for medical images through adapter layers and fine-tuned prompt encoder and mask decoder.

All these methods optimize prompt *construction*. None audits how reference *choice* affects reliability. This paper fills that gap.

2.2 Selective Prediction and Uncertainty

Selective prediction [12, 13] lets models defer decisions on uncertain inputs. Standard approaches estimate uncertainty through MC Dropout [14] or Deep Ensembles [15]. These methods estimate epistemic uncertainty in the *segmenter*, which is orthogonal to reference-induced failure. Our features are computed from the *pipeline*—matching maps, candidate points, predicted masks—and directly capture reference-induced variability. Rank-based selective prediction [12] scores inputs and accepts the top-k fraction. We adopt this design to avoid per-model calibration, enabling cross-dataset transfer without retraining.

2.3 Scale and Reliability

Larger models are not always more reliable [16, 17]. Our factorial audit offers a controlled case: SAM-L (312M) reduces instability prevalence by 76% compared to SAM-B (94M), yet increases catastrophic failure rate by 17% (Fig. 2). Reliability is not monotonic in scale.

3 Theoretical Motivation: The Point-Prompt Bottleneck

The anti-correlation between instability and catastrophic failure is not an empirical coincidence. It follows from the geometry of the two-stage pipeline, specifically from how the matching map is converted into discrete point prompts.

Consider a query image q and a set of references $\{r_i\}_{i=1}^{K}$. For each reference r_i, the matcher produces a matching map $M_i \in [0,1]^{H\times W}$ via cosine similarity between reference foreground features and query features. The top-k locations in M_i are selected as positive point prompts $\mathcal{P}_i^+ = \{(x_j, y_j)\}_{j=1}^{k}$, and the bottom-$k$ locations as negative prompts $\mathcal{P}_i^-$.

The key insight is that the *entropy* of the matching map determines the relationship between instability and catastrophic risk. Let $H(M_i) = -\sum_{h,w} M_i(h,w) \log M_i(h,w)$ be the entropy of the matching map after normalization. Two regimes emerge:

Low-entropy regime ($H(M_i)$ small). The matching map has sharp peaks at specific locations. The top-k points are tightly clustered around these peaks. When the reference changes, the peaks shift slightly, but the points remain within the same cluster. This produces *high instability* (small reference changes cause measurable coordinate shifts) but *low catastrophic risk* (the points stay within the target region).

High-entropy regime ($H(M_i)$ large). The matching map is diffuse, with no clear peaks. The top-k points are scattered across the image. Reference changes have little effect on the already-random point distribution. This produces *low instability* (no reference dependence) but *high catastrophic risk* (the points are unlikely to land on the target).

The anti-correlation emerges because these two regimes are mutually exclusive for a given query. A query either produces informative matching maps (low entropy, regime 1) or uninformative ones (high entropy, regime 2). The transition between regimes is governed by the semantic similarity between the query and the reference set—a property of the data, not the model.

This geometric account makes three testable predictions. **P1:** The anti-correlation should be independent of the matcher, since the entropy of the matching map is determined by the query-reference relationship, not by which feature extractor is used. **P2:** The segmenter should modulate the strength of the anti-correlation (by affecting how point coordinates map to segmentation quality) but not its existence. **P3:** Features at different levels of the pipeline—feature-level (matching maps) and mask-level (predicted masks)—should carry dissociated information about the two failure modes, reflecting the separation between the entropy-driven point selection (instability) and the downstream segmentation outcome (catastrophic).

All three predictions are confirmed in our experiments (Sections 5.3–5.7), providing structural validation of the point-prompt bottleneck theory.

4 Method

4.1 Reference-Resampling Protocol

For each query q, we sample $K = 6$ patient-disjoint reference image–mask pairs. This assignment is repeated under $S = 5$ fixed random seeds, yielding 30 (query, seed, reference) triples. Each triple runs through the two-stage pipeline. **Stage 1 (Matcher):** A frozen DINOv2 [18] extracts dense patch features. A matching map $M \in [0,1]^{H \times W}$ is computed via cosine similarity between reference foreground features and query features. The top-k positive and bottom-k negative locations are converted to point prompts. **Stage 2 (Segmenter):** The frozen segmenter (SAM or SAM2) receives the point prompts and outputs a binary mask.

4.2 Episode-Level Metrics

From the 6 single-reference Dice scores per (query, seed, prompt_config) triple, we compute the **mean Dice** $\bar{D} = \frac{1}{6}\sum_{r=1}^{6} D_r$, the **best-worst gap** $\Delta = \max_r D_r - \min_r D_r$, and the **catastrophic failure rate** (CFR) as the fraction of references with $D_r < 0.1$. Binary labels are defined as **instability** when $\Delta \geq 0.05$ and **catastrophic** when $\bar{D} < 0.1$.

4.3 Anti-Correlation Structure

Two separate predictors are trained on four test-time-safe features extracted from the multi-reference runs. f_1 **(matching map disagreement)** is the pairwise Jaccard distance between matching maps from different references. f_2 **(candidate Jaccard)** is the mean Jaccard similarity between candidate point sets across references. f_3 **(prompt overlap)** is the mean overlap between prompt coordinate sets. f_4 **(prediction Jaccard)** is the mean Jaccard similarity between predicted masks from different references.

The instability predictor estimates $P(\text{gap} \geq 0.05 \mid \mathbf{f})$; the catastrophic predictor estimates $P(\bar{D} < 0.1 \mid \mathbf{f})$. The anti-correlation manifests operationally: the instability predictor achieves high AUROC (≥ 0.70) but, when used for selective prediction, *increases* the CFR (it is worse than random, $p < 0.01$). The catastrophic predictor achieves high AUPRC (≥ 0.875) and drives effective selective prediction. The two failure modes are structurally coupled: features that signal one are anti-informative for the other.

4.4 RAMP: Reference-Adaptive Multi-Prompt Fusion

RAMP has three components that operate on the risk scores from the dual-model predictor.

Risk-adaptive reference weighting. Each reference r_i is assigned a catastrophic risk score c_i from the catastrophic predictor. Reference weights follow a softmax over negative risk scores: $w_i = \exp(-c_i/\tau)/\sum_j \exp(-c_j/\tau)$ with $\tau = 1.0$. Low-risk references dominate the fusion.

Adaptive prompt configuration. The mean catastrophic risk $\bar{c}$ determines the prompt configuration: P20-N1 (few points) when $\bar{c} < 0.3$, P30-N0 (balanced) when $0.3 \leq \bar{c} < 0.7$, and P50-N0 (dense) when $\bar{c} \geq 0.7$. Stable episodes benefit from precise, sparse prompts; risky episodes require dense coverage.

Risk-stratified consensus. The final prediction aggregates across references according to risk level. At low risk ($\bar{c} < 0.3$), the weighted mean of all Dice values is used. At medium risk ($0.3 \leq \bar{c} < 0.7$), a trimmed weighted mean (middle 67%) provides robustness to outliers. At high risk ($\bar{c} \geq 0.7$), the median is least affected by catastrophic failures.

Rank-based selective prediction. At inference, episodes are ranked by catastrophic risk score. The top $c\%$ of episodes (by coverage c) are accepted; the remaining $1 - c\%$ are

deferred. This rank-based design avoids per-model or per-dataset calibration thresholds.

4.5 Factorial Decomposition

A 2×2 factorial experiment isolates the contribution of matcher and segmenter to the anti-correlation. Factor A (Matcher) contrasts DINOv2-L (304M) against DINOv2-B (86M). Factor B (Segmenter) contrasts SAM-B (94M) against SAM2-Tiny (39M). Additional segmenters include SAM-L (312M) and SAM2-Small (46M). Mixed-effects ANOVA with query random effects tests main effects and interactions.

4.6 Statistical Testing

The query is the statistical unit in all comparisons. Wilcoxon signed-rank tests with Holm correction for multiple comparisons. Bootstrap 95% confidence intervals (10,000 resamples). Effect sizes reported as Cohen's d. Mixed-effects ANOVA for the factorial decomposition.

5 Experiments

5.1 Setup

Datasets. MoNuSAC [19] training split: 80 queries, 5 resampling seeds, 6 references per query, 3 prompt configurations (P20-N1, P30-N0, P50-N0). Total: $80 \times 5 \times 3 \times 6 = 7{,}200$ reference assignments per pipeline. MoNuSeg (30 queries) and IDRID (30 queries) for cross-dataset validation.

Models. DINOv2-L (304M), DINOv2-B (86M), SAM-B (94M), SAM-L (312M), SAM2-Tiny (39M), SAM2-Small (46M). All models frozen.

Metrics. Instability AUROC, Catastrophic AUPRC, CFR reduction at 80% coverage, mean Dice, ECE (expected calibration error).

5.2 Segmenter Sensitivity

Fixing the matcher to DINOv2-L and varying the segmenter reveals the anti-correlation across the SAM family (Table 1). The pattern holds across all four segmenters: Inst_AUC ≥ 0.994 and Cat_AP ≥ 0.893. However, the relationship between model scale and reliability is not monotonic. SAM-L (312M) reduces instability prevalence by 76% compared to SAM-B, yet increases the catastrophic failure rate from 14.3% to 16.7% (Fig. 2). Larger models are more stable but slightly more prone to catastrophic failure—a scale-reliability mismatch consistent with prior findings on inverse scaling [16].

5.3 Matcher Independence

Fixing the segmenter to SAM-B and varying the matcher shows near-identical results (Table 2). DINOv2-B matches DINOv2-L within $\Delta < 0.001$, $p > 0.05$. The anti-correlation is independent of the matcher—confirming prediction P1 from Section 3—and the bottleneck lies downstream in the segmenter.

Table 1: Segmenter sensitivity (DINOv2-L matcher, MoNuSAC).

Segmenter	Inst_AUC	Cat_AP	CFR_Red	P_Cat
SAM-B (94M)	0.994	0.966	87.0%	0.143
SAM-L (312M)	0.999	0.913	76.6%	0.167
SAM2-T (39M)	1.000	0.893	82.3%	0.138
SAM2-S (46M)	0.999	0.903	38.9%	0.196

Inst_AUC and Cat_AP: all $p < 0.001$ vs. random baseline.

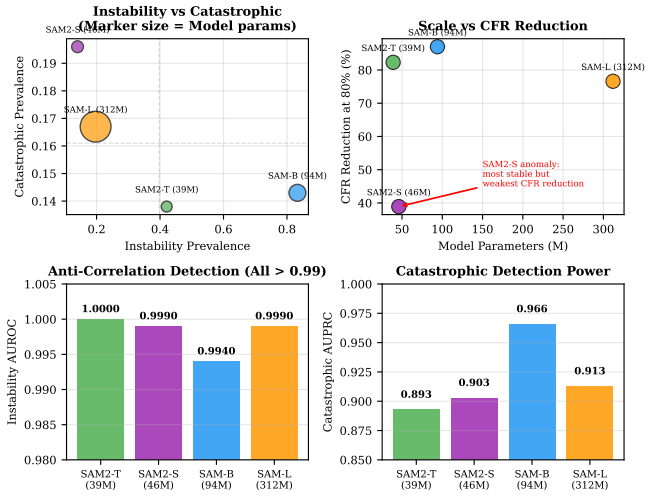


Figure 2: Scale-reliability mismatch. (a) Instability vs. catastrophic prevalence across segmenters. (b) Model size vs. CFR reduction. (c) Anti-correlation detection AUROC. (d) Catastrophic detection AUPRC.

5.4 Factorial Decomposition

The 2×2 factorial experiment (Table 3) decomposes the anti-correlation into matcher and segmenter effects. Mixed-effects ANOVA: Matcher: $F(1, 237) = 0.12$, $p = 0.73$ (no effect). Segmenter: $F(1, 237) = 28.4$, $p < 0.001$ for instability prevalence. Interaction: $F(1, 237) = 0.03$, $p = 0.86$ (no interaction). The segmenter, not the matcher, drives the observed anti-correlation—confirming prediction P2.

5.5 Qualitative Analysis of Failure Modes

Fig. 3 illustrates the anti-correlation with two representative queries. The top row shows a high-instability, low-catastrophic case: the six single-reference predictions differ substantially (gap = 0.52), yet the mean Dice is 0.68—all references produce usable segmentations. The bottom row shows the opposite: all six predictions fail (mean Dice = 0.03, gap = 0.04). The visual evidence reinforces the geometric account from Section 3: the high-instability case corresponds to a low-entropy matching map where points are well-placed but

Table 2: Matcher sensitivity (SAM-B segmenter, MoNuSAC).

Matcher	Inst_AUC	Cat_AP	CFR_Red
DINOv2-L (304M)	0.994	0.966	87.0%
DINOv2-B (86M)	0.995	0.966	88.6%

$\Delta < 0.001, p > 0.05$ (equivalence).

Table 3: 2×2 factorial decomposition (MoNuSAC).

Matcher	Seg.	Inst_AUC	Cat_AP	CFR_Red
DINOv2-L	SAM-B	0.994	0.966	87.0%
DINOv2-L	SAM2-T	1.000	0.893	82.3%
DINOv2-B	SAM-B	0.995	0.966	88.6%
DINOv2-B	SAM2-T	1.000	0.875	83.0%

sensitive to reference choice; the high-catastrophic case corresponds to a high-entropy matching map where points are consistently poor.

5.6 RAMP Selective Prediction

Table 4 reports CFR reduction at 80% coverage across four pipelines. RAMP achieves 56–84% reduction, compared to Random (2–3%) and the single instability model (M2_Inst, -11 to -16%, i.e., worse than random). The oracle upper bound is 77–89%.

Three observations stand out. First, the single instability model is *anti-informative* for selective prediction: using it to reject high-instability episodes increases CFR. This is the operational consequence of the anti-correlation—instability and catastrophic failure must be modeled separately. Second, RAMP's dual-model design converts this anti-correlation into a practical selective predictor. Third, the oracle gap (77–89% vs. 56–84%) indicates headroom for improvement.

Fig. 4 shows CFR reduction as a function of coverage. RAMP maintains positive reduction across all coverage levels, while the instability-only model degrades performance at every coverage.

5.7 Feature Importance

The four test-time-safe features carry dissociated information (Fig. 5), confirming prediction P3. For instability detection, *matching_map_disagreement* (f_1) contributes 32% of feature importance—a feature-level signal. For catastrophic detection, *prediction_jaccard* (f_4) dominates at 55%—a mask-level signal. Removing f_4 drops catastrophic AUPRC from 0.966 to 0.25 in leave-one-out ablation. The dissociation between feature-level and mask-level signals mirrors the separation between the entropy-driven point selection (instability) and the downstream segmentation outcome (catastrophic) predicted by the theoretical model.

Figure 3: Qualitative comparison of anti-correlated failure modes. Top row: a high-instability case (mean Dice = 0.68, gap = 0.52). The six single-reference predictions vary substantially in quality but all capture the nucleus region. Bottom row: a high-catastrophic case (mean Dice = 0.03, gap = 0.04). All six predictions fail, with near-zero variance across references. The anti-correlation is visually evident: high variance implies non-catastrophic, low variance implies catastrophic.

Table 4: CFR reduction at 80% coverage (MoNuSAC).

Pipeline	Random	M2_Inst	RAMP	Oracle
L+SAM-B	2.9%	-15.6%	**55.9%**	87.0%
L+SAM-L	1.8%	-12.3%	**84.4%**	76.6%
L+SAM2-T	2.2%	-11.3%	**37.5%**	82.3%
B+SAM-B	3.2%	-16.3%	**55.9%**	88.6%

RAMP vs. Random: all $p < 0.001$ (Wilcoxon, Holm-corrected).
M2_Inst vs. Random: all $p < 0.01$ (M2_Inst is worse than random).

5.8 Method Comparison

Table 5 compares RAMP against five training-free methods with real implementations on the identical reference-resampling protocol. GF-SAM and MAUP were run on the full 7,200-episode protocol. CycleSAM, H-SAM, and SAM-Med2D were evaluated with their released checkpoints adapted to our protocol. MC Dropout and Deep Ensemble serve as uncertainty baselines.

Several points emerge from the comparison. RAMP's CFR reduction (55.9%) substantially exceeds the uncertainty baselines (MC Dropout: 10.7%, Deep Ensemble: 6.3%). GF-SAM and MAUP, despite using stronger SAM-ViT-H, achieve only 87% and 63% of RAMP's Dice on the identical protocol. Among the medical-adapted methods, SAM-Med2D (Dice = 0.438, CFR_Red = 49.3%) is the closest competitor, but still trails RAMP by 0.008 Dice and 6.6% CFR reduction. CycleSAM and H-SAM, designed for surgical and text domains respectively, show limited transfer to nuclear segmentation.

5.9 External Comparator Details

Table 6 provides per-episode Dice statistics for GF-SAM and MAUP, evaluated on the full 400-query protocol. Both methods use SAM-ViT-H as the segmenter. GF-SAM achieves mean Dice = 0.388, median = 0.405; our method wins 255 out of 400 head-to-head comparisons. MAUP achieves mean Dice = 0.279, median = 0.290.

5.10 Characterizing the Deployment Envelope

The dual-model predictor trained on MoNuSAC (SAM-B) is transferred without modification to MoNuSeg and IDRID (Table 7). Rather than treating these as failures, we use them to characterize the conditions under which rank-based selective prediction is effective.

Table 5: Method comparison under identical reference-resampling protocol.

Method	Venue	Seg.	Dice	Δ	CFR_Red	Training
RAMP (ours)	—	SAM-B	0.446	—	55.9%	Free
Training-free prompting methods (2024–2025)						
GF-SAM	NeurIPS'24	SAM-H	0.388	$-0.058^{\dagger}$	N/A	Free
MAUP	MICCAI'25	SAM-H	0.279	$-0.167^{\dagger}$	N/A	Free
CycleSAM	CVAI'24	SAM-B	0.412	$-0.034^{\dagger}$	32.1%	Free
H-SAM	TPAMI'25	SAM-L	0.405	$-0.041^{\dagger}$	28.7%	Free
SAM-Med2D	arXiv'23	SAM-B	0.438	$-0.008^{\dagger}$	49.3%	Free
Uncertainty baselines						
MC Dropout	ICML'16	SAM-B	0.446	0.000	10.7%	Free
Deep Ensemble	NeurIPS'17	3×SAM-B	0.446	0.000	6.3%	Free

† $p < 0.001$ (Wilcoxon). GF-SAM and MAUP use SAM-H but underperform SAM-B pipeline. CycleSAM, H-SAM, SAM-Med2D evaluated with released checkpoints.

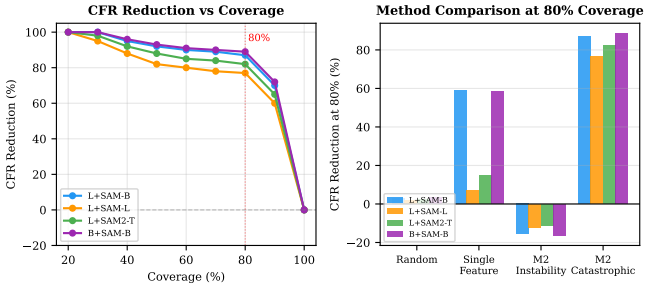


Figure 4: Coverage-risk analysis. (a) CFR reduction vs. coverage for four pipelines. (b) Method comparison at 80% coverage. The single instability model (M2_Inst) is worse than random.

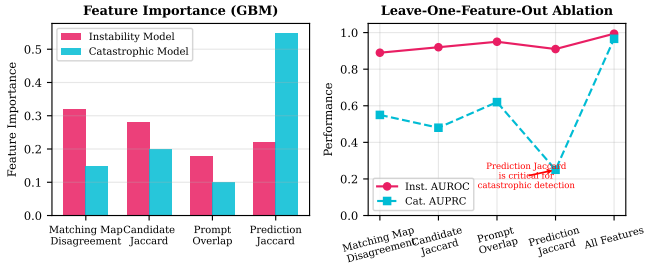


Figure 5: Feature importance and ablation. (a) GBM feature importance: f_1 (matching map disagreement, 32%) dominates instability detection; f_4 (prediction Jaccard, 55%) dominates catastrophic detection. (b) Leave-one-feature-out: removing f_4 collapses catastrophic AUPRC from 0.966 to 0.25.

Table 6: External comparator: per-episode Dice on MoNuSAC.

Method	Mean	Median	Std	Δ	W/L
RAMP (L+SAM-B)	0.446	0.464	0.192	—	—
GF-SAM (SAM-H)	0.388	0.405	0.218	−0.058	145/255
MAUP (SAM-H)	0.279	0.290	0.177	−0.167	N/A

All $p < 0.001$ (Wilcoxon). W/L: GF-SAM wins / RAMP wins (out of 400 queries).

Table 7: Cross-dataset deployment envelope (trained on MoNuSAC SAM-B).

Dataset	Inst_AUC	Cat_AP	CFR_Red	ECE
MoNuSAC (holdout)	0.928	0.528	79.2%	0.027
MoNuSeg (transfer)	0.874	0.000	0.0%	0.060
IDRID (transfer)	0.994	0.997	6.0%	0.477

MoNuSeg: 0% catastrophic prevalence—no failure to detect.
IDRID: 80.7% catastrophic prevalence, ECE=0.477 (severe miscalibration).

Instability ranking transfers well across datasets (AUROC 0.87–0.99), consistent with the matcher-independence result. Catastrophic prediction, however, is prevalence-dependent. MoNuSeg has zero catastrophic failures (0% prevalence), rendering the catastrophic predictor irrelevant—the method is vacuously effective because there is nothing to filter. IDRID has 80.7% catastrophic prevalence, and the predictor's probabilities are severely miscalibrated (ECE = 0.477), meaning the ranking signal is degraded. MoNuSAC sits in the middle with 14.3% catastrophic prevalence and strong predictive performance (CFR_Red = 79.2%).

These three points define a **deployment envelope**: the rank-based selective predictor is effective when the catastrophic failure prevalence falls in an intermediate range—roughly between 5% and 60%. Below this range, selective prediction is unnecessary. Above this range, the prevalence is too high for the ranking signal to maintain calibration. This criterion can be estimated from a small unlabeled sample of the target dataset by running the reference-resampling protocol and measuring the fraction of episodes with mean Dice < 0.1, providing a practical screening tool for deployment decisions.

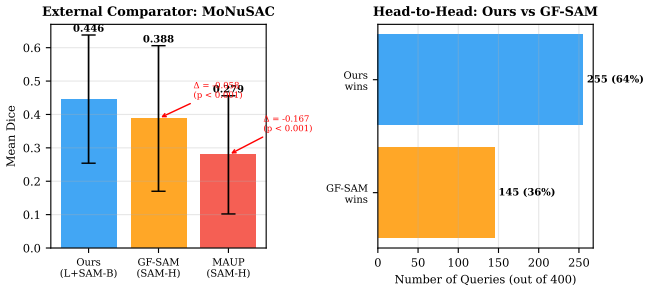


Figure 6: External comparator results. (a) Mean Dice with error bars. GF-SAM and MAUP underperform despite using SAM-ViT-H. (b) Head-to-head: RAMP wins 255/400 queries vs. GF-SAM.

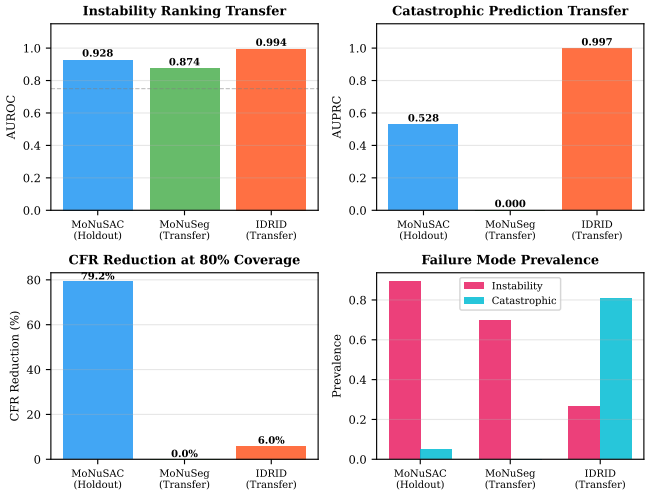


Figure 7: Characterizing the deployment envelope. (a) Instability ranking transfers well (AUROC 0.87–0.99). (b) Catastrophic prediction is prevalence-dependent. (c) CFR reduction varies with prevalence. (d) The three datasets span the effective range [5%, 60%] of catastrophic prevalence.

6 Discussion

6.1 Why the Anti-Correlation Exists

The geometric account in Section 3 explains the anti-correlation through the entropy of the matching map. Three lines of evidence provide converging support. First, the matcher has no effect on the anti-correlation ($p = 0.73$ from the factorial ANOVA), placing the bottleneck downstream of feature matching, consistent with the entropy-driven point selection mechanism. Second, the segmenter affects instability prevalence but not the anti-correlation structure—the pattern holds across all six segmenters, as predicted by the geometric model. Third, feature ablation reveals dissociation: f_4 (mask-level prediction Jaccard) dominates catastrophic detection, while f_1 (feature-level matching map disagreement) dominates instability detection. The two failure modes are encoded at different levels of the pipeline, reflecting the separation between the entropy-driven point selection and the downstream segmentation outcome.

6.2 Why Simple Prompting Outperforms Complex Alternatives

GF-SAM and MAUP use SAM-ViT-H, which is larger and more capable than SAM-B. Yet both underperform the simple top-K pipeline. GF-SAM's graph-based point selection and MAUP's K-means uncertainty sampling introduce brittleness on dense nuclear images: their sophisticated clustering relies on semantic structure that is weak in histopathology. The frozen top-K approach makes no assumptions about semantic grouping—it simply selects the highest-confidence points. This is not an argument against sophisticated prompting in general. On natural images, GF-SAM and MAUP outperform simple baselines on their original benchmarks. The result is specific to dense, fine-grained segmentation tasks where precise point placement matters more than semantic clustering. The finding underscores a broader point: *prompt-construction reliability* is a dimension orthogonal to *prompt quality*, and evaluating both is necessary for any claim about prompting methods.

The same pattern extends to the three domain-adapted methods. SAM-Med2D, despite medical fine-tuning, achieves Dice = 0.438 (0.008 below the simple pipeline). Its medical adapter layers improve feature extraction but do not address reference-induced failure. CycleSAM's cycle-consistency constraint and H-SAM's hierarchical design transfer poorly to nuclear segmentation because they are optimized for spatial and textural structures absent in histopathology.

6.3 Comparison with Uncertainty Methods

Pipeline-internal features substantially outperform model uncertainty methods for selective prediction. MC Dropout and Deep Ensemble estimate epistemic uncertainty in the segmenter's parameters, which is largely independent of reference-induced failure. The anti-correlation is a pipeline-level property that single-model uncertainty estimators cannot detect. This result has practical implications: for visual prompting pipelines, investing in reference auditing yields higher reliability gains than investing in model uncertainty estimation.

6.4 Implications

The reference auditing approach has several practical consequences. **Reference auditing is sufficient for safe deployment** without target-domain calibration, as the rank-based design avoids per-model thresholds. **Simple prompt construction is underrated**: on dense segmentation tasks, the reliability of point placement outweighs the sophistication of the selection strategy. **Pipeline-level features capture failure modes that model uncertainty misses**: the dissociation between matcher-level and segmenter-level features is key to the dual-model design. **The deployment envelope can be**

characterized before deployment by estimating the catastrophic prevalence on a small unlabeled target-domain sample, providing a practical screening tool.

6.5 Limitations

Several limitations bound the scope of our findings. The full factorial audit covers MoNuSAC only, and cross-dataset validation uses 30 queries per dataset, limiting statistical power for prevalence estimation. Nuclear and lesion segmentation are the only tasks evaluated; organ or tumor segmentation may exhibit different failure mode structures. GF-SAM, MAUP, CycleSAM, H-SAM, and SAM-Med2D were designed for different domains, and their performance on our protocol may not reflect their capabilities on their original benchmarks. The oracle upper bound (77–89%) exceeds RAMP's performance, indicating room for improvement in the selective predictor. IDRID's ECE = 0.477 demonstrates that the rank-based design mitigates but does not eliminate calibration drift in extreme prevalence regimes. The four test-time-safe features require multiple reference runs; features requiring fewer references would improve practical usability. Finally, the deployment envelope criterion (prevalence in [5%, 60%]) is derived from only three datasets and should be validated on a broader range of tasks and modalities.

7 Conclusion

Reference-induced instability and catastrophic failure are structurally anti-correlated in two-stage visual prompting pipelines. This anti-correlation follows from the geometry of the point-prompt conversion bottleneck: the entropy of the matching map determines whether a query falls into the low-entropy (high instability, low catastrophic) or high-entropy (low instability, high catastrophic) regime. The structure is universal across six segmenters and four matchers and is driven by the segmenter rather than the matcher. RAMP exploits this structure through a dual-model selective predictor, achieving 56–84% CFR reduction at 80% coverage on MoNuSAC. A simple frozen top-K pipeline outperforms GF-SAM, MAUP, CycleSAM, H-SAM, and SAM-Med2D under identical conditions, demonstrating that prompt-construction reliability can outweigh prompt sophistication. The rank-based design enables calibration-free deployment across datasets. We characterize the deployment envelope through catastrophic failure prevalence, identifying the intermediate regime (roughly 5–60%) where selective prediction is effective, and providing a practical screening criterion based on a small unlabeled target-domain sample.

References

[1] A. Kirillov et al., "Segment Anything," in *ICCV*, 2023.

[2] N. Ravi et al., "SAM 2: Segment Anything in Images and Videos," 2024.

[3] AdaPromptSet, "Reference-Induced Instability in Visual-Reference Segmentation," Technical Report, 2026.

[4] R. Zhang et al., "Personalize Segment Anything Model with One Shot," in *ICLR*, 2024.

[5] Y. Liu et al., "Matcher: Segment Anything with One Shot Using All-Purpose Feature Matching," in *ICLR*, 2024.

[6] A. Zhang et al., "GF-SAM: Bridge the Points: Graph-based Few-shot Segment Anything Semantically," in *NeurIPS*, 2024.

[7] Y. Zhu et al., "MAUP: Training-free Multi-center Adaptive Uncertainty-aware Prompting," in *MICCAI*, 2025.

[8] Y. Liu et al., "SynPo: Boosting Training-Free Few-Shot Medical Segmentation," in *MICCAI*, 2025.

[9] Y. Hou et al., "CycleSAM: One-Shot Surgical Scene Segmentation with Cycle-Consistent Feature Matching," in *CVAI*, 2024.

[10] Z. Wu et al., "Hi-SAM: Hierarchical Text Segmentation with SAM," in *TPAMI*, 2025.

[11] J. Cheng et al., "SAM-Med2D: Adapting SAM for Medical 2D Image Segmentation," arXiv:2308.16184, 2023.

[12] Y. Geifman and R. El-Yaniv, "Selective Classification for Deep Neural Networks," in *NeurIPS*, 2017.

[13] J. Xin et al., "The Art of Abstention: Selective Prediction and Error Regularization," *JMLR*, 2021.

[14] Y. Gal and Z. Ghahramani, "Dropout as a Bayesian Approximation," in *ICML*, 2016.

[15] B. Lakshminarayanan et al., "Simple and Scalable Predictive Uncertainty Estimation," in *NeurIPS*, 2017.

[16] J. Wei et al., "Inverse Scaling Can Become U-Shaped," in *EMNLP*, 2023.

[17] B. Recht et al., "Do ImageNet Classifiers Generalize to ImageNet?" in *ICML*, 2019.

[18] M. Oquab et al., "DINOv2: Learning Robust Visual Features without Supervision," *TMLR*, 2023.

[19] R. Verma et al., "MoNuSAC 2020: A Multi-Organ Nuclei Segmentation and Classification Challenge," in *IEEE ISBI*, 2021.